Run report: $k = 5$ Edwards route with F4 trace replay, 2026-10-15T17:23Z to 2026-10-20T03:58Z

Descriptive record of the runs; observations only. The interpretation is in `../../REPORT_EDWARDS.pdf` and `../../README.md`. Rendered from `summary_edwards.json` (a copy of the study's file, beside this report).

Spec. `PROTOCOL_EDWARDS.md`, frozen at commit `3a8599b3f`. Orchestration `run_edwards.sh`, then `run_edwards_resume.sh` after the resumable driver was added. Binary `examples/joux_vitse_edwards.rs`. The four cost runs, `p = 31` seed 1, `p = 61` seed 1 and `p = 31` seed 2 ran the build of `3a8599b3f` (no checkpointing). `p = 41` seed 2, and the counted runs of `p = 71` seed 1 and `p = 41` seed 1, ran the resumable build, saved as the frozen binary (SHA-256 `622a47d3`…, built 2026-10-16T03:39Z at commit `c2cb903ac`). The trial, trace, accounting and records are the same between the two builds; only checkpointing and the wall-time cap were added (`DEVIATION.md` items 4 and 7).

Host. Claude Code cloud container, 4 vCPU Intel Xeon @ 2.10 GHz, 15 GB, rustc 1.97.0. Isolation: none. Arms overlapped on the host at times and the host was restarted at least three times.

Timeline (UTC). Cost runs 17:23 on 2026-10-15. DLP arms: `p = 31` seed 1 17:24 to 18:14 on the 15th; `p = 61` seed 1 20:16 on the 15th to 00:02 on the 16th; `p = 31` seed 2 02:14 to 04:07 on the 16th; `p = 41` seed 2 04:07 to 06:55 on the 16th; `p = 71` seed 1 from 06:55 on the 16th, resumed at 11:52 on the 16th and 16:12 on the 19th, solved 17:57 on the 19th; `p = 41` seed 1 counted run 17:57 on the 19th to its cap at 03:58 on the 20th. Two earlier attempts at `p = 41` seed 1 and `p = 71` seed 1 were killed by a restart at about 02:05 on the 16th and their partial records kept apart (`killed_20261016T0205Z/`, not used). Every deviation, with its reason, is in `data/edwards/DEVIATION.md`.

Requested and completed. Cost: 4 runs × 32 residuals, all complete. DLP: 6 arms requested; 5 solved with both answers verified; 1 (`p = 41` seed 1, a six-point base) stopped at its 10-hour wall cap with 3 of 7 relations. No arm was substituted.

Measured (unit: `F_p` multiplications; counts, not wall time). Checks failed: **0**.

record	$\lvert F\rvert$	state	trials	relations	IC online	rho $\mathbb{F}_p$ mult. (ops)
dlp_p31_s1.jsonl	7	solved, verified	105344	8	$5.93{\cdot}10^{12}$	$1.75{\cdot}10^{6}$ (3868)
dlp_p31_s2.jsonl	6	solved, verified	310048	7	$1.24{\cdot}10^{13}$	$2.15{\cdot}10^{6}$ (4765)
dlp_p41_s1.jsonl	6	stopped at the cap	1377312	3 of 7	$8.76{\cdot}10^{13}$	$5.8{\cdot}10^{6}$ (12832)
dlp_p41_s2.jsonl	8	solved, verified	534016	9	$2.07{\cdot}10^{13}$	$1.07{\cdot}10^{7}$ (23658)
dlp_p61_s1.jsonl	14	solved, verified	451408	15	$1.68{\cdot}10^{13}$	$1.36{\cdot}10^{7}$ (30071)
dlp_p71_s1.jsonl	12	solved, verified	1833280	13	$6.71{\cdot}10^{13}$	$1.29{\cdot}10^{7}$ (28580)

cost record	residuals	committed solve, mean	replay, mean	factor, mean	checks failed
cost_p271_s1.jsonl	32	$5.85{\cdot}10^{7}$	$3.39{\cdot}10^{7}$	1.74	0
cost_p271_s2.jsonl	32	$5.85{\cdot}10^{7}$	$3.31{\cdot}10^{7}$	1.77	0
cost_p521_s1.jsonl	32	$5.86{\cdot}10^{7}$	$3.4{\cdot}10^{7}$	1.74	0
cost_p521_s2.jsonl	32	$5.86{\cdot}10^{7}$	$3.32{\cdot}10^{7}$	1.77	0

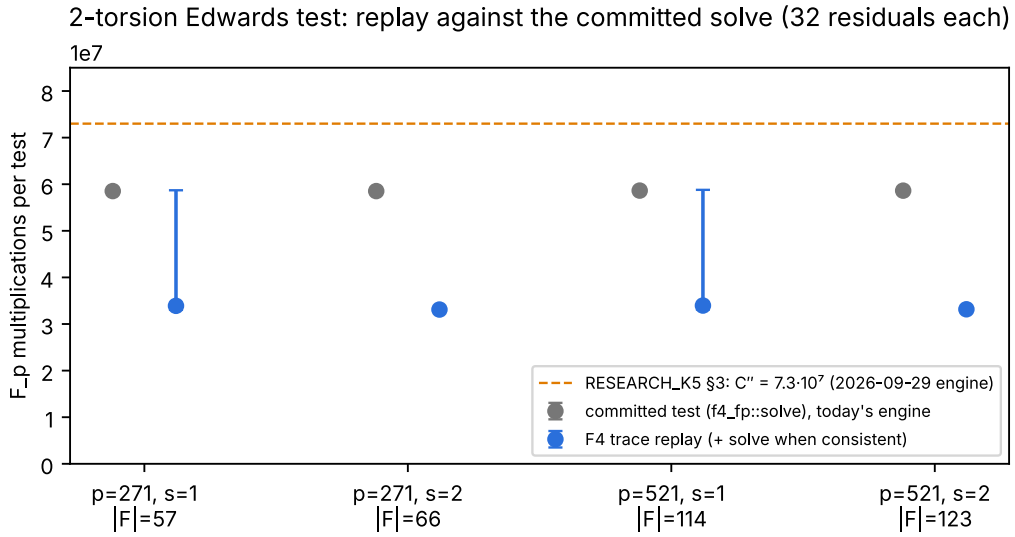


Figure 1: Cost of one test per residual, plotted from the cost records by `../../figures/plot_edwards.py`.